ODT Summative 2

Idea Brief :

Smart Locker System :

A multi-gateway security locker, encompassing physical hardware and a mobile application. The first gateway uses an OTP generated on the app, which you enter on the safe to unlock the door. The second gateway challenges you with a math equation, and its answer is the exact distance at which you place your hand for the sensor to detect it and open the gate. The third gateway uses a coloured pattern inside the safe to open another mini locker.

PHYSICAL STRUCTURE	DESCRIPTION
Main Locker	Primary enclosure with the core locking mechanism, push buttons, NeoPixel, buzzer, servo motor, door sensor, and ultrasonic sensor.
Mini Locker	Smaller compartment physically attached beside the main locker, with independent locking mechanism (dc motor), only accessible after the main locker is opened.
Back Wall (main locker)	The interior back wall of the main locker has 4 coloured sheets (physical, non-electronic) fixed to it, revealed only once the main door swings open, serving as the visual clue for the mini locker sequence.
Distance Scale	A physical scale placed in front of the locker, so that users can position their hand at the correct distance during Gateway 2.

Gateway 1 –OTP

CONCEPT

A **one-time password** is generated each session from the MIT App Inventor application. The user reads the OTP from the app and enters it physically on the locker using push buttons.

How It Works

- The app generates a randomised numeric OTP (within a defined range/digit count) each time the user requests one.
- The OTP is transmitted to the ESP32 over Bluetooth.
- The user enters the code digit-by-digit using push buttons labelled 1-9 on the locker face.
- The ESP32 compares the entered sequence against the received OTP in real time.

Push button Sound Feedback

Each button press triggers a short, distinct frequency tone from the buzzer, providing the user with audio confirmation of each digit entered.

COMPONENT	CORRECT ENTRY RESPONSE	WRONG ENTRY RESPONSE
NeoPixel	Lights up green (sequentially)	Lights up red (sequentially)
Servo Motor	Rotates to open position [PWM]	No response
Buzzer	**Door sensor** detects door physically swinging open, triggers buzzer at frequency 1	No door sensor response

Lockout Condition

After 3 consecutive wrong attempts, the entire Gateway 1 input system locks, the push buttons stop responding and no further entries are accepted.

Gateway 2 – Math Equation w/ Ultrasound sensor distance

CONCEPT

Instead of entering a code, the user must solve a math problem and position their hand at that distance from an ultrasonic sensor mounted on the locker.

How It Works

- The app generates a simple arithmetic equation whose answer always falls within a defined range, ideally **5 to 20**(e.g. "3 × 4 + 2 = ?", answer 14).

- The equation is displayed on the app interface.
- A physical scale or ruler is affixed to the locker beside the ultrasonic sensor, marked in centimetres.
- The user solves the equation mentally, then places their hand at the corresponding centimetre mark in front of the sensor.
- The ESP32 reads the ultrasonic distance.
- If correct → same success response as Gateway 1 (green NeoPixel, servo opens, door sensor triggers buzzer)
- If wrong → same rejection response (red NeoPixel, rejection buzzer, no servo)

Gateway 3 - Colour Sequence [Mini Locker]

CONCEPT

Can only be unlocked after the main locker has already been opened, using a colour sequence revealed on the interior back wall.

How It Works

- Once the main locker door opens (via Gateway 1 or 2), it reveals the interior back wall covered with 4 coloured sheets
- This sequence of colours is the "key" to the mini locker
- The MIT App interface contains buttons in the same colours as the wall, whose purpose, prior to the opening of the locker, was deliberately kept vague
- The user presses the coloured buttons in the same sequence as the colours on the wall
- Correct → mini locker servo opens
- Wrong → rejection response on NeoPixel and buzzer

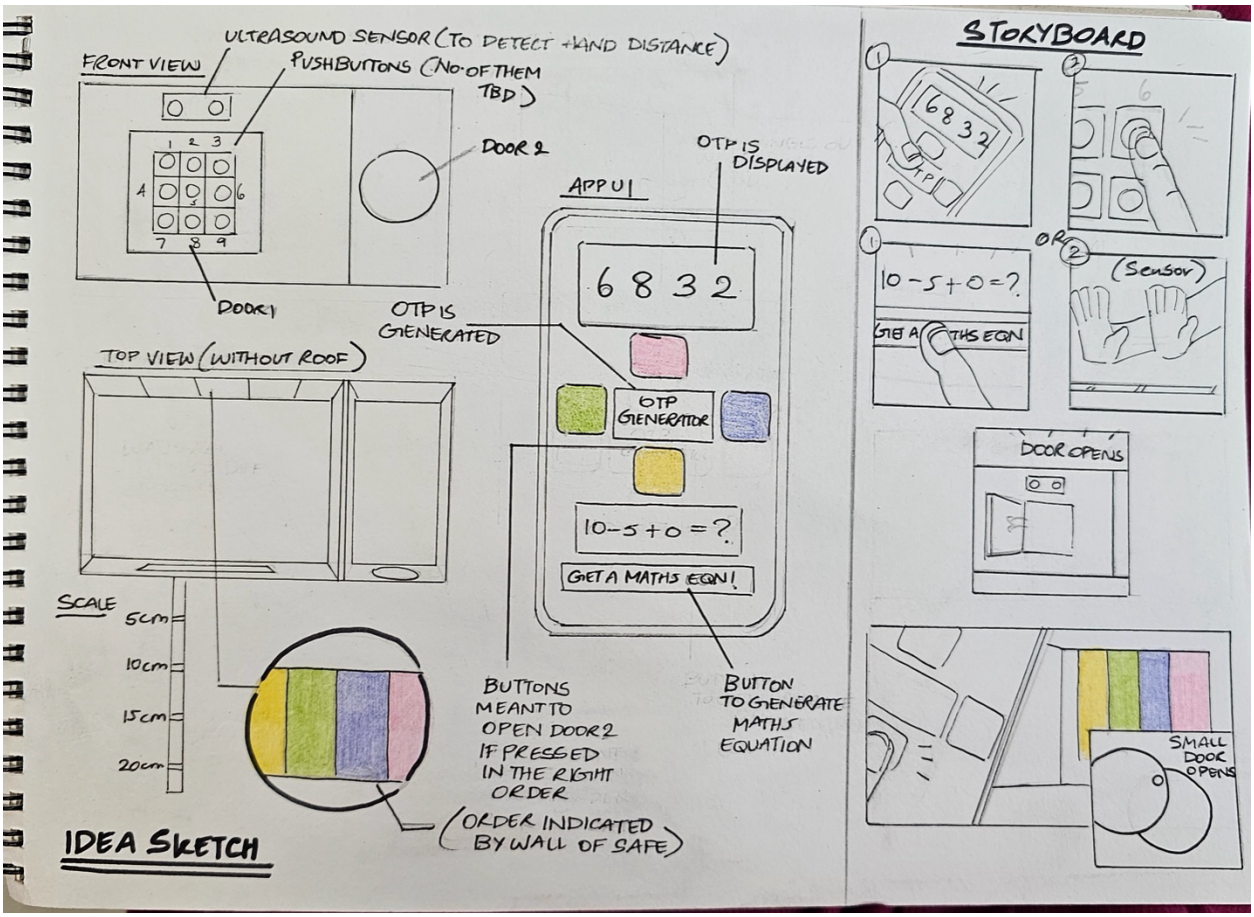
App Interface

COMPONENT	**FUNCTION**
OTP generation button	Random four digit OTP is generated when button is pressed to be entered into the safe
Maths equation generator	Random maths equation is generated, whose answer is the distance in cm at which the user's hand is to be kept for the Ultrasound sensor to sense it
Coloured buttons (4)	When pressed in the right order will unlock the mini locker
OTP Display	Displays the OTP generated on pressing the button
Equation Display	Displays the equation generated on pressing the button

How it works

When OTP button is pressed, the code **generates a random OTP** and sends it to the phone **via Bluetooth**, where it appears on the display. Entering this OTP using push buttons opens the door.

When pressing the equation generator button, an equation is generated and displayed. When this **equation is solved and your hand is placed at that distance**, the sensor senses it and the door of the main locker opens.

When the coloured buttons are pressed **according to the pattern on the wall inside the safe**, the mini locker door opens.

Basic Idea Sketch :